

MECHANICAL DESIGN & REPORT

Project Title: Bio-Inspired Quadruped Robot (Design Phase)

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Track: Systems Engineering & Industrial Mechanics

1. Executive Summary & Introduction

This technical report encapsulates the initial mechanical design, structural layout, and kinematic configurations of a bio-inspired quadruped robot dog. In robotics, failure is rarely confined to software boundaries; structural instabilities, poor actuator integration, excessive friction, and uncalculated torque requirements represent the predominant catalysts for mechanical failure. This document mathematically verifies the actuator stall margins, establishes the center of gravity optimization matrix, proposes a statically stable locomotion gait, and provisions the electronic component integration schema.

2. Chassis Layout & Structural Geometry

The main chassis (torso) is engineered utilizing a box-shaped parallelepiped geometry, optimized to maximize internal volume while maintaining structural stiffness. The layout isolates the electronic control layer from the heavy power distribution components.

- **Torso Dimensions:** Width = 40 mm, Length = 80 mm, Height = 20 mm.
- **Component Isolation:** The control board (microcontroller) and sensor stack are fixed on the upper deck, protecting them from electromagnetic interference and mechanical stress.
- **Power Subsystem Integration:** To optimize stability, the heavy battery pack and primary toggle switch are integrated directly into the bottom-most quadrant of the chassis.

Engineering Principle (Lowering CoG): Concentrating the primary mass (battery pack) at the lowest geometric baseline minimizes the moment arm of structural tipping, significantly lowering the overall Center of Gravity (CoG) and improving dynamic recovery margins during locomotion.

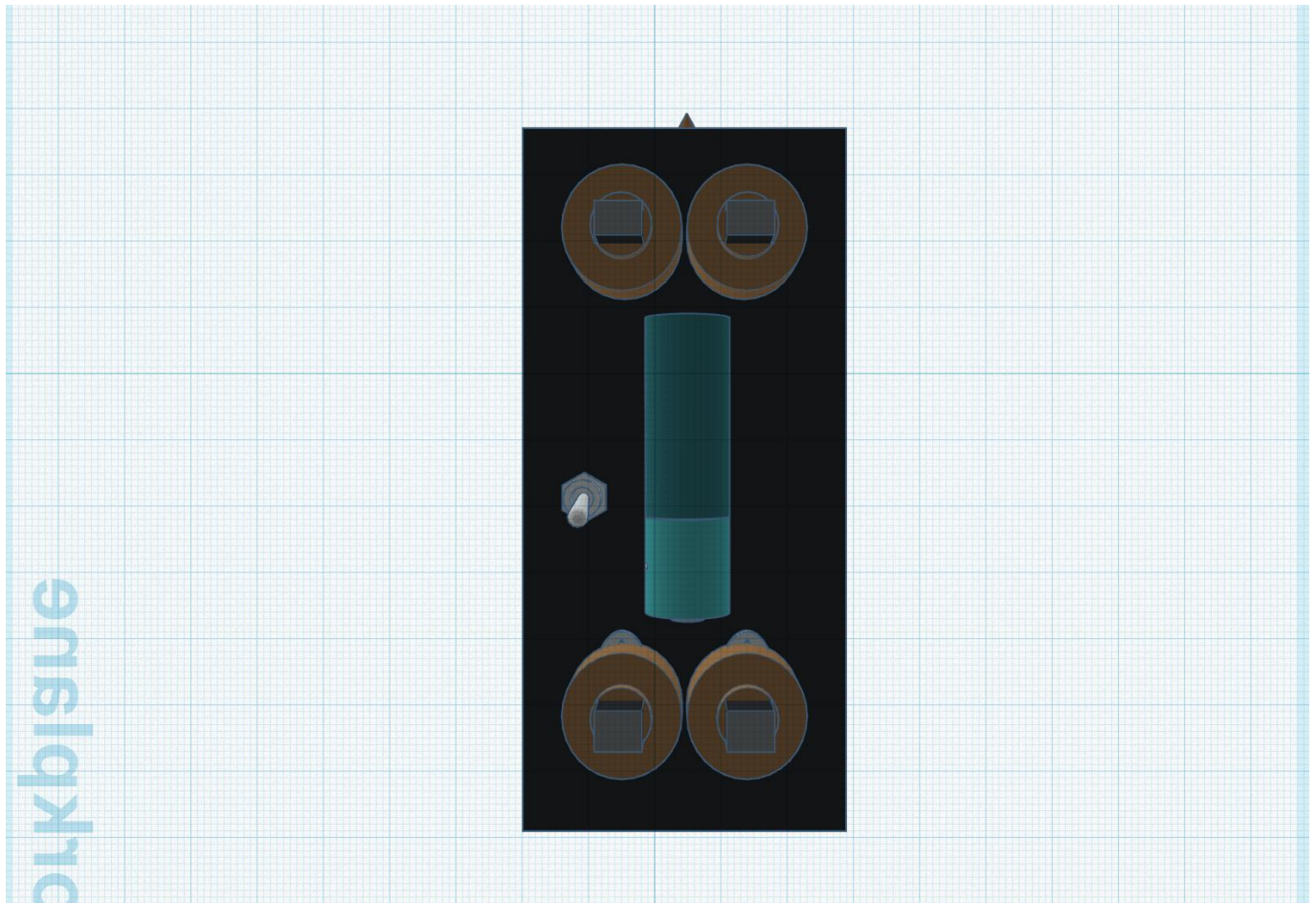


Figure 1: Side Profile and Center of Gravity Configuration Matrix

3. Leg Architecture & Kinematic Degrees of Freedom (DOF)

The robot adopts an insectoid leg configuration to distribute operational loads evenly. Each independent limb assembly is designated with two distinct rotational axes, providing high agility while keeping system complexity manageable.

Limb Assembly	Joint Identifier	Motion Profile	Degrees of Freedom (DOF)
4 x Symmetric Legs	Hip Joint (Abduction/Adduction)	Rotational movement along the pitch axis (Forward/Backward sweep)	1 DOF per leg
	Knee Joint (Flexion/Extension)	Rotational movement along the pitch axis (Vertical extension/retraction)	1 DOF per leg
Total Operational System Degrees of Freedom (DOF):			8 DOF

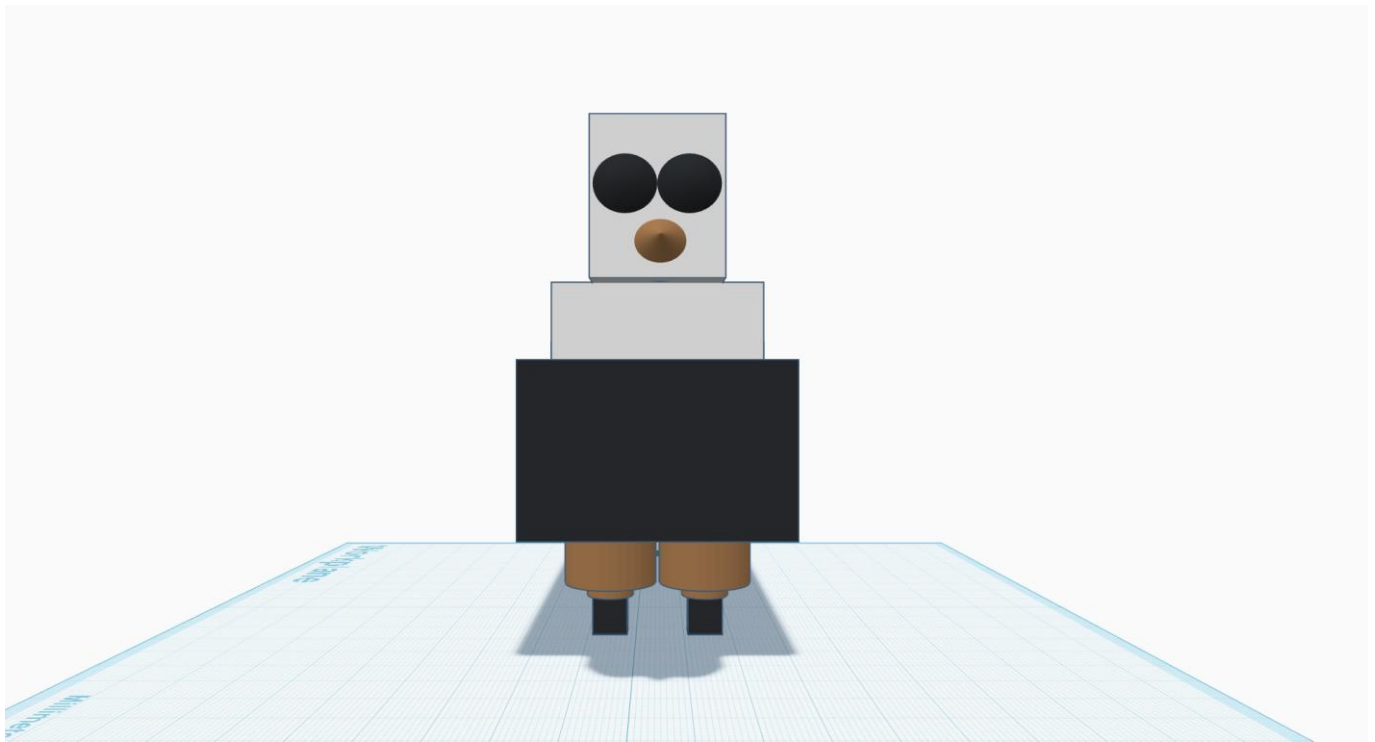


Figure 2: Front Orthographic Projections and Symmetric Actuation Axes

4. Actuator Selection & Electronic Architecture

To satisfy the structural demands of the 8 DOF kinematic model, precise actuator and sensor ecosystems are integrated into the design framework.

4.1 Actuator Selection: High-Torque Servo Motors

Standard DC motors provide high angular velocity but suffer from extremely low torque output, making them unsuitable for joint articulation. Therefore, the design utilizes **MG996R Metal Gear Servo Motors** for all 8 joints. These servos feature internal multi-stage gear reduction units that trade rotational speed for enhanced torque output. They also feature integrated potentiometer-driven feedback loops for precise closed-loop angular positioning.

4.2 Sensor Stack & Control Avionics

- **Microcontroller Node (The Brain):** An Arduino Nano or ESP32 board manages data processing and outputs PWM signals to the servo drivers.
- **Obstacle Avoidance Core:** An **HC-SR04 Ultrasonic Sensor** is embedded inside the front head assembly (represented visually by the dual ocular apertures on the Tinkercad head block). This sensor tracks forward distances to actively prevent collisions.
- **Balance & Orientation Module:** An **MPU6050 Inertial Measurement Unit (IMU)** is mounted inside the main torso to monitor pitch, roll, and dynamic tilting in real-time.

5. Theoretical Static Torque Calculations

To ensure the MG996R servo motors do not undergo thermal runaway or gear stripping, we calculate the worstcase static torque requirement for a single leg joint when the robot is holding its position.

5.1 Primary Variables & Assumptions

- Total estimated mass of the fully assembled robot (m) = **1.20 kg** (includes structural frame, 8 servos, battery, and sensors).
- Acceleration due to gravity (g) = **9.81 m/s²**.
- Maximum effective horizontal lever arm length from joint axis to load point (r) = **0.07 m** (7.0 cm).

5.2 Mathematical Formulation

When standing stationary, the total weight is assumed to be distributed evenly across all four limbs. Therefore, the static force (F) acting on a single limb is calculated as:

$$F = (m \times g) / 4$$
$$F = (1.20 \text{ kg} \times 9.81 \text{ m/s}^2) / 4 = 2.943 \text{ N}$$

The absolute minimum static torque (τ) required at the joint to maintain structural equilibrium against this force *static* is:

$$\tau_{static} = F \times r$$
$$\tau_{static} = 2.943 \text{ N} \times 0.07 \text{ m} = 0.206 \text{ Nm}$$

5.3 Factor of Safety (FoS) Allocation

Selecting an actuator based solely on static equilibrium will cause failure during actual movement due to joint friction, dynamic acceleration spikes, and mechanical backlash. A standard engineering Factor of Safety (**FoS = 2.5**) is applied:

$$\tau_{design} = \tau_{static} \times FoS$$
$$\tau_{design} = 0.206 \text{ Nm} \times 2.5 = 0.515 \text{ Nm}$$

Converting the design requirement into standard hobby-servo metrics ($1 \text{ Nm} \approx 10.197 \text{ kg}\cdot\text{cm}$):

$$\tau_{design} \approx 0.515 \times 10.197 \approx 5.25 \text{ kg}\cdot\text{cm}$$

Actuator Compliance Verification: The selected MG996R servo motor delivers a stall torque of **11.0 kg·cm** at an operational voltage of 6.0V. Since **11.0 kg·cm > 5.25 kg·cm**, the selected actuators possess a safe performance margin.

6. Stability Analysis & Support Polygon Metrics

The structural stability of a quadruped robot is governed by the relative positioning of the projection of its Center of Gravity (CoG) with respect to its support polygon (the geometric shape formed by linking the points where the feet touch the ground). By mounting the battery pack and heavy switch directly beneath the chassis centerline, the horizontal CoG stays centered inside the torso, maximizing balance during unexpected shifts.

7. Locomotion Strategy: Statically Stable Crawl Gait

To ensure smooth movement without tipping over, a **Statically Stable Crawl Gait** is chosen. This locomotion strategy relies on moving only one limb at a time while the remaining three limbs maintain full contact with the ground.

1. **Initial State:** All four limbs are planted on the ground, creating a rectangular support polygon.
2. **Step 1:** The front-right leg lifts and moves forward. The remaining three legs form a triangular support polygon that safely keeps the CoG within its borders.
3. **Step 2:** The rear-left leg lifts and sweeps forward to restore symmetry.
4. **Step 3:** The front-left leg lifts and steps forward.
5. **Step 4:** The rear-right leg lifts and steps forward, completing one full walking cycle.

This sequence guarantees that the robot always has three points of contact on the ground, allowing it to remain balanced at any moment in the walking cycle.

8. Risk Assessment & Anticipated Mechanical Challenges

During the upcoming physical prototyping and assembly phase, the following mechanical challenges are anticipated and must be monitored:

- **Mechanical Backlash:** Small gaps between the plastic and metal gears inside commercial servo motors can introduce slight looseness at the joints, causing body oscillations. This will be minimized using predictive software filtering.
- **High Joint Friction:** Friction at the 3D-printed plastic interfaces can increase the torque load on the servos. Incorporating low-friction washers or brass bushings at the pivot points will resolve this.
- **Limb Misalignment:** Tiny structural defects during assembly can cause the limbs to become uneven, leading to drifting during forward movement. This will be corrected using software trim calibration offsets.

